SHIVAM KAMBOJ

Ph.D. Candidate in Physics (Expected Fall 2026)

First-Principles Quantum Theorist · Open Quantum Systems · Computational Modeling & Hardware Control

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Research Profile: Theoretical and computational physicist specializing in building quantum models from the microscopic Hamiltonian up. Experienced in bridging symmetry-resolved sparse exact diagonalization, open-system Lindblad dynamics, and non-Markovian noise modeling with physical device realization. Proven track record closing the loop between theory and hardware via pulse-level qudit control and linear-inversion process tomography at **Lawrence Livermore National Laboratory (LLNL)**, experimental cavity-QED collaboration, and production-grade open-source QEC software (`nonmarkov_qec` on PyPI).

EDUCATION

University of California, Merced **Expected Fall 2026**
Ph.D. in Physics | Advisor: Prof. Lin Tian
Research Focus: Light-matter interactions, open quantum Rabi systems, parity-protected bosonic cat qubits, and superconducting circuit-QED modeling.

San Jose State University **2022**
M.S. in Physics | Advisor: Prof. Hilary M. Hurst
Research Focus: Analytical dispersion relations and numerical dynamics of 2D driven magnetic spin-torque oscillator arrays.

RESEARCH & NATIONAL LABORATORY EXPERIENCE

Tian Group, University of California, Merced **May 2023 – Present**
Graduate Student Researcher (Theoretical Quantum Optics & Open Quantum Systems) *Merced, CA*

- **First-Principles Hamiltonian Modeling:** Formulate theoretical and computational models of ultrastrong light-matter coupling in the quantum Rabi dimer, directly connecting microscopic Hamiltonian structure to spectra, collective normal modes, and macroscopic entanglement.
- **Symmetry-Resolved Sparse Exact Diagonalization:** Combine analytical Bogoliubov/normal-mode analysis with sparse exact diagonalization (CSR format, Krylov subspace eigensolvers) to resolve low-energy states by parity and site-exchange symmetry, distinguish true vs. avoided level crossings, and characterize a candidate parity-protected cat-state doublet.
- **Open-System Dynamics & HPC Scaling:** Simulate Lindblad master-equation state evolution, entanglement, and dissipative photon loss using Python and QuTiP; enforce rigorous Hilbert-space truncation convergence and execute checkpointed parameter studies on Linux clusters with SLURM.
- **Finite-Element Device Virtualization:** Build COMSOL RF and AC/DC electromagnetic models of a proposed superconducting implementation; extract capacitances, eigenmode frequencies, impedances, and inductive participation ratios to assess feasible interactions and physical device constraints.

Lawrence Livermore National Laboratory (LLNL) **May 2024 – Aug 2024**
Physics Graduate Intern, Quantum Coherent Device Physics Group *Livermore, CA*

- **Pulse-Level Qudit Control:** Developed modular Python software for pulse-level control of multilevel superconducting qudits ($|1\rangle \rightarrow |3\rangle$ transitions), evaluated optimal control gate sequences, and characterized hardware performance.
- **Linear-Inversion Quantum Process Tomography (LI-QPT):** Translated noisy experimental measurements into reconstructed process maps through a generalized linear-inversion tomography pipeline using Moore-Penrose pseudo-inverse reconstruction.
- **Automated Coherence & Calibration Fitting:** Automated Ramsey interferometry and T_1/T_2 relaxation fitting routines to characterize device drift and deliver rapid quantitative feedback for closed-loop control experiments.

Sharping Group, University of California, Merced **2024 – Present**
Research Collaborator, Experimental Quantum Optics & Superconducting Realization *Merced, CA*

- **Hybrid Cavity-Magnon Modeling:** Developed first-principles Hamiltonian models of coupled YIG-cavity systems to study magnon-photon spectra and entanglement, including inter-cavity coupling and parametric driving.
- **Theory-to-Hardware Translation:** Collaborate directly with experimentalists at postdoc-level autonomy to trans-

late theoretical Rabi dimer specifications into fabricable superconducting circuit architectures.

Hurst Group, San Jose State University and UC Merced **Jun 2021 – Jan 2024**
Graduate Researcher, Driven Dissipative Spin Systems *San Jose / Merced, CA*

- **Non-Hermitian Spin-Torque Arrays:** Derived analytical dispersion relations and developed numerical solvers for two-dimensional spin-torque nano-oscillator arrays; studied oscillatory edge modes and robustness to hardware disorder, leading to a first-author publication in *Physical Review B*.

SCIENTIFIC SOFTWARE & PRODUCTION PROJECTS

`nonmarkov_qec` — **Open-Source Surface-Code QEC Benchmarking Library** **2026 – Present**
Creator & Maintainer | Released on PyPI (Python 3.11+, Stim, PyMatching, NumPy/SciPy, GitHub Actions CI)

- **End-to-End Correlated Noise Pipeline:** Built a surface-code benchmarking pipeline combining stochastic Ornstein-Uhlenbeck and sum-of-OU $1/f$ noise generation, fast compiled-template Stim noise injection, PyMatching MWPM decoding, and two-layer Monte Carlo sampling with parametric-bootstrap 95% confidence intervals.
- **Matched-Marginal Physics Isolation:** Compared temporally correlated dephasing against a statistically matched Markovian white-noise baseline across code distances $d = 3$ and $d = 5$ to isolate temporal memory effects from marginal error-rate drift.
- **Production Software Rigor:** Maintained 74 automated unit and analytic-validation tests (`pytest`), strict static type checking (`mypy --strict`), linting (`ruff`), continuous integration, and automated PyPI releases with design documentation.

Physics Concept Analyzer Tool (PCAT) — Cloud-Deployed AI Application **2025**
Lead Developer | Python, Flask, Google Gemini API, Google Cloud Platform (App Engine) *UC Merced*

- Designed and deployed a full-stack physics reasoning application on Google App Engine integrating multimodal LLM workflows to guide users through concept identification, physical assumptions, and model construction before calculation.

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

1. **S. Kamboj**, R. A. Duine, B. Flebus, and H. M. Hurst, "Oscillatory edge modes in two-dimensional spin-torque oscillator arrays," *Physical Review B* **109**, 094436 (2024). `doi:10.1103/PhysRevB.109.094436`.
2. **S. Kamboj** and L. Tian, "Deterministic Cat States and Parity-Protected Bosonic Qubits in the Quantum Rabi Dimer." *Manuscript in preparation* (2026).
3. **S. Kamboj**, J. Beurkens, J. Sharping, and L. Tian, "Superconducting realization of the Quantum Rabi Dimer." *Manuscript in preparation* (2026).

SELECTED PRESENTATIONS & INVITED TALKS

- **S. Kamboj** and L. Tian, "The Instability Window: Macroscopic Entanglement and Delayed Symmetry Breaking in the Quantum Rabi Dimer." *Quantum Industry Showcase, UC Santa Barbara* (2026).
- **S. Kamboj** and L. Tian, "Adiabatic Evolution and Fock-state Mapping Across the Phase Boundary in a Quantum Rabi Dimer." Poster, *APS Global Physics Summit* (2026).
- **S. Kamboj**, Y. Cho, and Y. J. Rosen, "Improving Fidelity of Quantum Gates Using Optimal Control." Poster, *PLS Division Student Poster Symposium, Lawrence Livermore National Laboratory* (2024).
- **S. Kamboj**, "Simulating Quantum Systems, and Quantum Key Distribution." *Invited Talk, Friday Seminar Series, CSU Stanislaus* (2024).
- **S. Kamboj** and H. M. Hurst, "Topology of a Driven 2D Spin Torque Oscillator Array." *APS March Meeting, Chicago*, Abstract T00.249 (2022).

TECHNICAL SKILLS

Domain	Core Methods, Tools & Technologies
Theory & Open Systems	Microscopic Hamiltonian modeling; Lindblad master equations; non-Markovian & $1/f$ correlated noise; light-matter & magnon-photon coupling; Bogoliubov & normal-mode analysis; parity-resolved spectra; sparse exact diagonalization; Fock-space convergence analysis.
QEC & Data Analysis	Surface-code QEC benchmarking (Stim, PyMatching); linear-inversion process tomography (LI-QPT); pseudo-inverse reconstruction; Ramsey & T_1/T_2 fitting; two-layer Monte Carlo simulation; parametric bootstrap uncertainty estimation.
Scientific Software & HPC	Python, NumPy, SciPy, QuTiP, Qiskit, Stim, PyMatching, Matplotlib; Linux, Bash, SLURM HPC clusters, Git, GitHub Actions CI/CD, `pytest`, `ruff`, `mypy --strict`; Flask, Google Gemini API, GCP.
Device Modeling & Control	COMSOL RF and AC/DC finite-element modeling; capacitance, impedance & inductive participation extraction; superconducting qudit pulse control and process characterization.

TEACHING, HONORS & ACADEMIC LEADERSHIP

- **Teaching & Mentoring:** Guest Lecturer, Computational Physics (PHYS 281), UC Merced (2024, 2026); Teaching Assistant in Physics, UC Merced & San Jose State University (2021–Present); Mentor, Learning Assistant Program (2022).
- **Honors & Fellowships:** Outstanding Graduate Teaching Award, UC Merced (2026); Physics Graduate Group Summer Fellowship, UC Merced (2025, 2026); Physics Graduate Group Travel Fellowship, UC Merced (2026); Outstanding Teaching Assistant Award, SJSU (2022); 1st Place, Tech-Invent Silicon Particle Detector Competition (2017).
- **Leadership & Service:** Physics Delegate, Graduate Student Association, UC Merced (2024); Member, Graduate Dean Advisory Council on Diversity, UC Merced (2023).